

Page-1: Title Page

- **Project Title:** Customer Segmentation Project
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Page-2: Objective

Objective

To analyze customer purchasing behavior and segment customers using clustering techniques.

Page-3: Tools & Process

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn

Process

Loaded dataset

Selected customer income and spending score

Applied KMeans clustering

Visualized customer groups

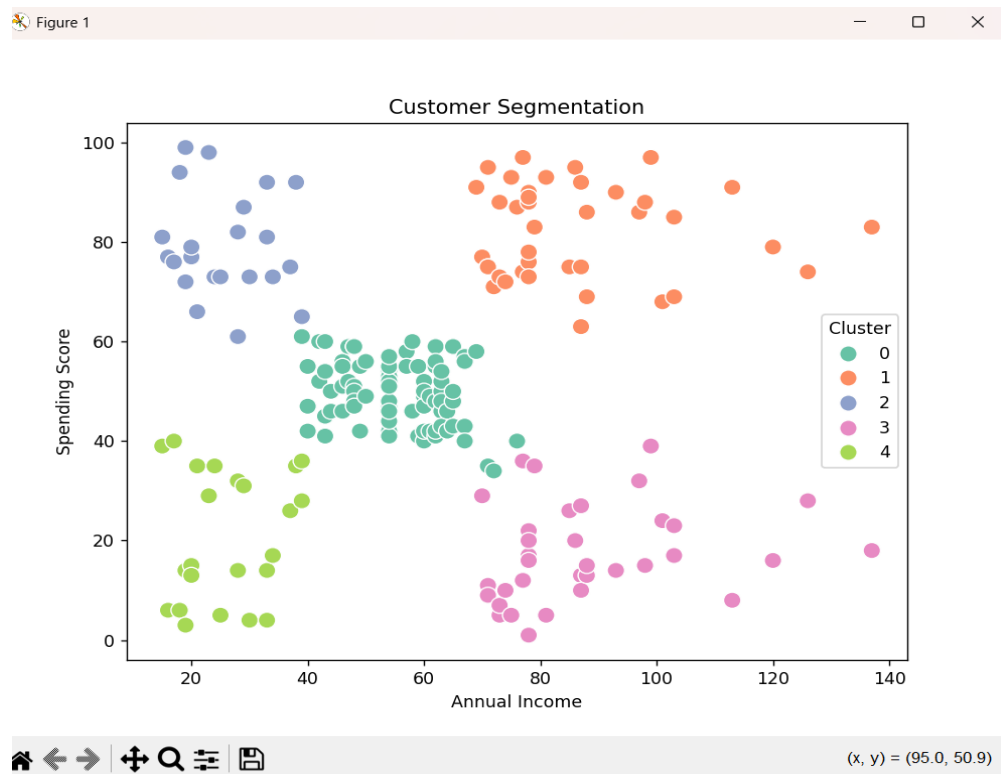
Page-4: Output

Output

The screenshot shows a VS Code editor with a file named `customer_segmentation.py` open. The script imports `pandas`, `matplotlib.pyplot`, and `seaborn`, then uses `sklearn.cluster.KMeans` and `sklearn.preprocessing.StandardScaler` to process data from `dataset/Mall_Customers.csv`. The terminal output shows the execution of the script, displaying the first 5 rows of the dataset and the results of the KMeans clustering.

```
customer_segmentation.py
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import seaborn as sns
4
5 from sklearn.cluster import KMeans
6 from sklearn.preprocessing import StandardScaler
7
8 # Load dataset
9 data = pd.read_csv("dataset/Mall_Customers.csv")
10
11 # Display first 5 rows
12 print(data.head())
13
```

```
PS C:\Users\sainy\OneDrive\Desktop\Customer_Segmentation_Project> python customer_segmentation.py
CustomerID Gender Age Annual Income (k$) Spending Score (1-100)
0 1 Male 19 15 39
1 2 Male 21 15 81
2 3 Female 20 16 6
3 4 Female 23 16 77
4 5 Female 31 17 40
CustomerID Gender Age Annual Income (k$) Spending Score (1-100) Cluster
0 1 Male 19 15 39 4
```



Conclusion

Customer_Segmentation_Report.pdf

